# The 28th Chinese Physics Olympiad Finals Problems

## I. (15 points)

Fix a smooth vertical semicircular rail at A、B, diameter AB＝$2R$; AB is 60° from vertical. A small smooth ring on it has mass $m$. A light thin smooth elastic cord of stiffness $k$ passes through it, its ends tied at A、B. Directly below A, at C, the cord is unstretched. Release it from rest at C. At $t$, it reaches C'; C'O makes angle $\theta$ with radius OB. Gravity is $g$. Find the rail force magnitude for the cases （1）$\theta$＝90°;（2）$\theta$＝30°.

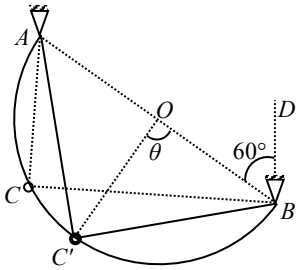


Fig. 28 Final—1

## II. (15 points)

A small cart, mass $M$, length $L$, is on a level floor. Two springs are fixed near the bottom at both ends, collinear, with rest lengths $l_1$, $l_2$ and stiffness $k_1$, $k_2$. The other ends meet small balls of masses $m_1$、$m_2$, unattached. Initially ball 1 compresses spring 1, holding the system at rest. Ball 2 is locked to the cart floor. Ball 2 is spring 2's rest length from the right end. Release ball 1; when spring 1 regains rest length, unlock ball 2 at once. Balls 1 and 2 collide and stick instantly. From release of ball 1 to spring 2's maximum compression, the cart moves $l_3$. No friction. Find initial spring 1 length $l$ and spring 2's peak compression $\Delta l_2$.

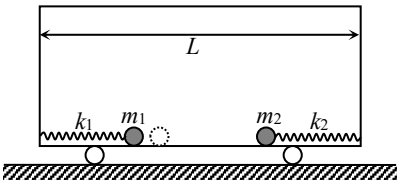


Fig. 28 Final—2

## III. (20 points)

Station A circles Earth at radius $r_A$＝$6.73\times10^6$m. Satellite B circles in the same plane, with radius $r_B$＝$\frac{3}{2}r_A$. A and B move counterclockwise. Launch a ship （no station recoil） to collect it. For fuel economy, except brief acceleration or braking, it coasts on identical anticlockwise transfer ellipses both ways. Both lie in the orbital plane. Perigee is on the station orbit, apogee on the satellite orbit. Given Earth radius $R_e$＝$6.38\times10^6$m and surface gravity $g$＝9.80m/s$^2$, find:

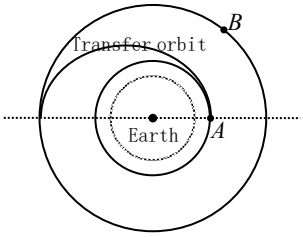


Fig. 28 Final—3

（1） The departure increment $\Delta v_A$ from station A to the ellipse. If at apogee it meets satellite B, find capture speed increment $\Delta v_A$ for zero relative motion.

（2） The least time to collect it and return, and the least angle station A must turn around Earth.

## IV. (15 points)